NexSight AI

Manufacturing Intelligence Platform

Microsoft Build AI Hackathon 2025

Theme: AI Meets Data — From Noise to Insight

Repository: github.com/KavinKumar-A/NexSight_AI

The Problem

- **Manufacturing lines generate a flood of noisy telemetry**
  Thousands of sensor readings per hour across temperature, vibration, humidity, speed and power — most of it never analysed.
- **Defects are expensive and discovered too late**
  PCB faults (opens, shorts, mousebites...) slip through manual inspection; root causes stay hidden in the data.
- **Insight is buried under raw numbers**
  Operators see dashboards of metrics, not answers. Quality patterns, failure risks and corrective actions go unnoticed.
- **Goal**
  Turn raw, noisy shop-floor data into clear, explainable, actionable manufacturing intelligence.

Our Solution — NexSight AI

- **One platform, noise to insight**
  Ingests sensor telemetry + real PCB inspection imagery and surfaces what actually matters.
- **Discovers hidden patterns**
  Statistical correlation mining across shifts, machines and product lines (validated $p<0.01$).
- **Explains root causes**
  Feature-importance / SHAP decomposition of every quality driver.
- **Predicts and prevents**
  XGBoost forecasts defects, yield and machine-failure risk before they happen.
- **Sees defects**
  Computer-vision PCB defect detection with Grad-CAM visual explanations.
- **Acts in real time**
  Live WebSocket sensor stream + SSE alerts + an AI assistant for natural-language Q&A.

Theme Alignment — From Noise to Insight

- **NOISE → raw inputs**
  10,000 synthetic telemetry records + 1,500 real DeepPCB image pairs (10,013 annotations).
- **SIGNAL → AI engines**
  Pattern discovery, anomaly detection, root-cause, prediction and computer vision.
- **INSIGHT → outputs**
  Health score, ranked recommendations, failure forecasts and live alerts.
- **ACTION → decisions**
  Prioritised, ROI-ranked corrective actions an operator can execute today.

Key Features

- **Real-time streaming**
  WebSocket sensor feed + Server-Sent-Event alert stream.
- **Pattern discovery engine**
  Surfaces 7+ hidden quality correlations automatically.
- **Root-cause intelligence**
  SHAP / permutation-importance explainability.
- **Predictive analytics**
  XGBoost models for defect, yield and failure risk.
- **Computer-vision inspection**
  DeepPCB defect detection + Grad-CAM heatmaps.
- **Health score + anomaly detection**
  Composite KPI and Isolation-Forest anomaly flags.
- **AI recommendations + assistant**
  Ranked actions and natural-language data Q&A.

Architecture & Tech Stack

- **Backend**
  FastAPI (Python 3.12) — single service exposing REST, WebSocket and SSE.
- **Machine learning**
  scikit-learn, XGBoost, LightGBM; SHAP for explainability.
- **Computer vision**
  OpenCV + Pillow with Grad-CAM; works with or without PyTorch.
- **Frontend**
  Zero-build vanilla JS + Plotly.js + Canvas API single-page dashboard.
- **Data layer**
  Synthetic telemetry generator + real DeepPCB dataset + cached analytics.
- **Delivery**
  `pip install -r requirements.txt` then `python run.py` — one command to run.

AI / ML Approach

- **Pattern discovery**
  Correlation + statistical testing reveals shift / machine / environment effects.
- **Root cause**
  Model feature importance and SHAP attribute defects to concrete drivers.
- **Prediction**
  XGBoost regression/classification forecasts yield and failure risk (R^2 ~0.87).
- **Computer vision**
  Defect localisation from DeepPCB annotations; Grad-CAM shows model focus.
- **Anomaly detection**
  Isolation Forest + domain thresholds flag excursions in real time.
- **Explainability first**
  Every insight ships with a confidence score and the 'why' behind it.

Data Strategy

- **Real data — DeepPCB**
  1,500 PCB image pairs, 10,013 defect annotations across 6 defect classes.
- **Synthetic data — engineered**
  10,000 telemetry records with deliberately embedded quality patterns.
- **Why both**
  Real data grounds the CV; synthetic data lets us prove pattern-discovery works.
- **Live data**
  Each session adds streaming WebSocket readings, mixed into the metrics.
- **Reproducible**
  Data and trained models are committed; caches pre-warm on startup.

Impact & Results

- **1,535 defects detected**
  Across 200 PCB inspection images with full type + severity breakdown.
- **7 hidden patterns surfaced**
  Validated with $p<0.01$ statistical significance.
- **Failure risk forecasting**
  Flags high-risk machines 48-72 hours ahead for preventive action.
- **Quantified savings**
  Recommendation engine projects measurable yield gain and cost reduction.
- **Sub-2s alerts**
  Threshold breaches surface in real time over WebSocket / SSE.

Live Demo

- **Executive dashboard**
  Health score, KPIs and live machine status at a glance.
- **Defect intelligence**
  CV results, severity breakdown and defect encyclopedia.
- **Patterns & predictions**
  Discovered correlations and forecast charts.
- **AI assistant**
  Ask questions in plain English and get explained answers.
- **How to run**
  pip install -r requirements.txt → python run.py → http://localhost:8000

Roadmap

- **Train the CNN end-to-end**
  Replace annotation-driven CV with a trained PCBDefectNet + full Grad-CAM.
- **Azure integration**
  Deploy on Azure App Service; wire Azure OpenAI into the assistant.
- **Streaming ingestion**
  Connect real PLC / IoT Hub sensor feeds in place of the simulator.
- **Automated actioning**
  Push recommendations directly to maintenance / work-order systems.

Thank You

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